

Myopia Risk Prediction System

Nency Pansuriya, Unnati Tank
Dr. Jalpesh Vasa, Asst. Prof. Mukti Patel

Department of Artificial Intelligence and Machine Learning, Chandubhai. S. Patel Institute of Technology, Charotar University of Science and Technology Anand (Gujarat), India

Abstract. Myopia, commonly known as nearsightedness, is an increasingly prevalent vision disorder among children and adolescents worldwide. The condition is primarily influenced by modern lifestyle changes, including prolonged screen exposure, excessive near-work activities such as reading and mobile usage, reduced outdoor time, and genetic predisposition. If not detected early, myopia can progress into high myopia, leading to severe complications such as retinal detachment, glaucoma, and permanent vision impairment. Early identification of individuals at risk is critical for implementing preventive strategies and slowing disease progression. However, traditional diagnostic methods rely heavily on clinical equipment and professional examination, making them less accessible in schools, rural areas, and low-resource environments. To address this challenge, this paper presents a Myopia Risk Prediction System, an AI-powered web-based application designed to estimate the likelihood and progression of myopia using easily obtainable, non-invasive input parameters. These parameters include demographic details (age, gender), lifestyle factors (screen time, outdoor activity, near-work habits), physical attributes (height, weight), and genetic indicators (family history and parental myopia status). The models are implemented using advanced algorithms such as XGBoost and Gradient Boosting, ensuring high predictive accuracy and robustness. The backend is developed using Flask, while the frontend utilizes React for an interactive and user-friendly interface. The system provides real-time predictions, visual risk representation through an interactive dashboard, and personalized recommendations such as reducing screen time, increasing outdoor exposure, and seeking medical consultation when necessary. Overall, the proposed solution serves as a non-invasive, cost-effective, and scalable screening tool that can be deployed in schools, communities, and remote areas. It demonstrates the potential of artificial intelligence in preventive healthcare by enabling early detection and intervention, ultimately reducing the long-term burden of myopia.

Keywords: Myopia Risk Prediction, Machine Learning, XGBoost, Gradient Boosting, Risk Assessment, Refractive Error Detection, Preventive Healthcare, Web Application, Artificial Intelligence

1 Introduction

1.1 Background

Myopia, also known as nearsightedness, is one of the most common vision disorders in children worldwide. The condition often develops during school age and can worsen over time if not detected early. Increasing screen exposure, reduced outdoor activity, and genetic factors have contributed to the rising prevalence of myopia. Because many children do not visit an eye specialist until symptoms become noticeable, there is a need for an accessible screening system that can identify risk early. This project addresses that need by creating an AI-based myopia risk prediction system that estimates the likelihood of myopia based on simple input data such as age, screen time, outdoor activity, near-work habits, family history, and other lifestyle indicators.

1.2 Limitations of Conventional Eye Examination Methods

Traditional eye examinations for detecting conditions such as myopia rely heavily on specialized ophthalmic equipment, controlled clinical environments, and trained medical professionals, which makes them less accessible for large-scale or routine screening, especially in schools and rural or low-resource areas. Due to these limitations, many children do not undergo regular vision check-ups, and early signs of refractive errors often go unnoticed until the condition has already progressed. This delay in diagnosis leads to a higher risk of worsening vision, increased dependency on corrective measures, and potential long-term complications. Furthermore, the growing influence of modern lifestyle factors such as excessive screen time, prolonged near-work activities, and reduced outdoor exposure has significantly accelerated the prevalence of myopia, making early detection even more critical. Therefore, there is a pressing need for a cost-effective, accessible, and non-invasive screening system that can identify at-risk individuals at an early stage and enable timely preventive intervention.

1.3 Vision Need for the System

There is a strong need for an effective system to address the increasing cases of myopia, especially among children and adolescents. Current screening methods are often expensive, require specialized medical equipment, and are limited to clinical settings, making them inaccessible for large populations such as schools and rural communities. As a result, early detection is frequently missed, leading to the progression of the condition and increased risk of serious vision problems. Therefore, it is essential to develop a solution that provides early screening, is cost-effective, and can be easily accessed by a wide range of users. An AI-based system that uses simple input data and delivers quick, reliable predictions can help bridge this gap by enabling timely intervention and promoting preventive eye care.

1.4 Proposed Solution

The proposed system is an AI-powered solution designed to predict the risk of myopia using easily available, non-invasive input data such as lifestyle habits, demographic details, and family history. Unlike traditional methods, it does not require any specialized medical equipment or clinical setup, making it highly accessible and suitable for use in schools and remote areas. The system leverages advanced machine learning algorithms to analyze the input data and provide instant risk predictions along with a clear classification of the severity level. Additionally, it generates personalized preventive recommendations, such as reducing screen time, increasing outdoor activities, and seeking professional consultation when necessary. This approach ensures early detection, promotes awareness, and enables users to take proactive steps toward maintaining better eye health.

2 Problem Statement

2.1 Limitations of Existing Myopia Detection Systems

Previous approaches to myopia detection have largely focused on clinical examinations and vision testing methods that require specialized instruments such as autorefractors and Snellen charts. These systems, while accurate, are not suitable for large-scale deployment due to their dependency on healthcare infrastructure and trained professionals. Additionally, many existing digital health solutions focus only on basic symptom tracking rather than predictive analysis, limiting their effectiveness in early risk identification. Some research efforts have explored machine learning-based prediction models; however, they often rely on limited datasets or require clinical parameters that are not easily accessible to the general population. Furthermore, these systems lack real-time interaction and user-friendly interfaces, reducing their practical usability in everyday environments such as schools or homes. These limitations highlight the need for a more accessible and intelligent system capable of performing early screening using simple input data.

2.2 Challenges in Risk Prediction Using Non-Clinical Data

Predicting the risk of myopia using non-clinical data presents several challenges, particularly in terms of data reliability, feature selection, and model accuracy. Lifestyle-based inputs such as screen time, outdoor activity, and near-work habits are often self-reported, which may introduce inconsistencies and noise in the dataset. Additionally, identifying the most relevant features that significantly contribute to myopia development requires careful analysis and domain knowledge. Earlier models using traditional statistical approaches often failed to capture complex relationships between multiple risk factors, resulting in lower prediction accuracy. Moreover, ensuring that machine learning models generalize well across diverse populations remains a critical concern. Therefore, there is a need for robust machine learning techniques, such as XGBoost, that can handle heterogeneous data and provide reliable predictions even with limited or noisy inputs.

2.3 Need for Multi-Stage Prediction Systems

Most existing systems focus on single-stage prediction, typically identifying whether a person has myopia or not, without considering the progression or severity of the condition. Such approaches provide limited insights and are insufficient for preventive healthcare planning. A more comprehensive solution requires a multi-stage prediction pipeline that can detect refractive errors, estimate the risk of progression, and predict the severity in terms of diopter levels. Integrating multiple models, such as classification and regression techniques, enables a more detailed and interpretable analysis of the condition. The use of advanced ensemble learning methods like Gradient Boosting allows for improved prediction performance and better handling of complex data patterns. This multi-stage approach is essential for delivering actionable insights rather than simple binary outputs.

2.4 Need for User-Centric and Scalable Solutions

The effectiveness of any screening system depends not only on prediction accuracy but also on its usability and accessibility. Many existing systems lack intuitive interfaces and fail to present results in a manner that is easily understandable by non-expert users such as parents, teachers, or students. Additionally, the absence of personalized recommendations limits their practical value in promoting preventive actions. A scalable solution must provide real-time results, visual representations of risk, and actionable guidance for users. Technologies such as React and Flask enable the development of interactive and responsive web applications that enhance user experience. Therefore, there is a strong need for a system that combines accurate prediction with user-centric design, ensuring accessibility, scalability, and real-world applicability.

3 Materials and Methods

3.1 System Overview

The proposed system is designed as a multi-stage machine learning pipeline to accurately predict the risk and severity of Myopia using non-invasive and easily obtainable input features. The architecture consists of three primary layers: input acquisition, data processing and model inference, and output visualization. The system operates as a web-based application, where users provide relevant information through an interactive interface. This data is processed by the backend, which integrates trained machine learning models to perform sequential prediction tasks. Each stage of the pipeline contributes to a comprehensive risk assessment, including detection of refractive error, estimation of progression risk, and prediction of diopter severity. The modular design ensures scalability, maintainability, and efficient integration of additional features in future enhancements.

3.2 Data Collection and Features

The effectiveness of the proposed system depends on the quality and relevance of input

data, which is collected in the form of demographic, lifestyle, and genetic features. Demographic attributes include age and gender, which play a significant role in understanding the onset and progression patterns of myopia. Lifestyle-related factors such as daily screen time, duration of near-work activities, outdoor exposure, and participation in sports are included to capture behavioral influences that contribute to vision deterioration. Physical attributes such as height and weight are also considered to provide additional contextual information. Furthermore, genetic predisposition is incorporated through family history and parental myopia status, as hereditary factors are known to significantly increase the risk. These features are selected based on domain knowledge and existing research to ensure that the model captures both environmental and biological contributors to myopia development.

3.3 Data Preprocessing

Before being used for prediction, the collected data undergoes a series of preprocessing steps to ensure consistency, accuracy, and compatibility with machine learning models. Initially, data validation is performed to check for missing, inconsistent, or out-of-range values, which are either corrected or handled appropriately. Numerical features such as age, screen time, and outdoor activity duration are normalized to bring them into a common scale, thereby improving model performance and convergence. Categorical variables, including gender and parental myopia status, are transformed into numerical representations using encoding techniques such as label encoding or one-hot encoding. Additionally, feature engineering techniques may be applied to derive new attributes that enhance predictive power. These preprocessing steps are crucial for reducing noise, improving data quality, and ensuring that the models can effectively learn meaningful patterns from the input data.

3.4 Machine Learning Pipeline

The core of the system is a three-stage machine learning pipeline that performs sequential prediction tasks to provide a comprehensive assessment of Myopia risk. In the first stage, a classification model based on XGBoost is used to detect the presence of refractive error, providing an initial indication of potential vision problems. In the second stage, another classification model predicts the progression risk level, categorizing individuals into low, medium, or high-risk groups based on the input features. The third stage employs a regression model based on Gradient Boosting to estimate the diopter severity, offering a quantitative measure of vision impairment. The outputs from all three stages are combined using a rule-based scoring mechanism that integrates model predictions into a final risk score and classification. This multi-stage approach enhances prediction accuracy and provides more detailed insights compared to traditional single-stage models.

3.5 Output Generation

The final stage of the system focuses on presenting the prediction results in a clear, interpretable, and user-friendly manner. The processed outputs from the machine learning pipeline are converted into a risk percentage and categorized into predefined levels such as low, moderate, and high risk. These results are displayed on the frontend through interactive visual components, including risk gauges, summary cards, and graphical indicators, enabling users to easily understand their risk status. In addition to

risk classification, the system generates personalized recommendations tailored to the user's input data, such as reducing screen time, increasing outdoor activities, maintaining proper reading posture, and seeking professional eye examination when necessary. The output interface is designed to ensure accessibility for non-expert users, including parents, teachers, and students, thereby promoting awareness and encouraging preventive actions. This comprehensive output generation mechanism transforms complex model predictions into actionable insights, making the system effective for real-world deployment.

4 Technology Used

The proposed Myopia Risk Prediction System is developed using a robust and scalable technology stack that integrates web development frameworks, machine learning libraries, and deployment tools to ensure efficient performance and real-time interaction. The backend of the system is implemented using Flask, a lightweight and flexible Python-based web framework that facilitates seamless integration of machine learning models with RESTful APIs. Flask is responsible for handling user requests, performing data validation and preprocessing, and executing model inference to generate predictions.

On the frontend, the system utilizes React along with TypeScript and Vite to create a dynamic, responsive, and user-friendly interface. React enables efficient component-based development, allowing real-time updates and interactive visualization of results such as risk gauges and summary dashboards. This ensures that users can easily interact with the system and understand prediction outcomes without requiring technical expertise.

For the machine learning component, advanced ensemble learning algorithms such as XGBoost and Gradient Boosting are employed due to their high predictive accuracy and ability to handle complex, non-linear relationships in data. These models are implemented using libraries such as scikit-learn and are trained on structured datasets containing demographic, lifestyle, and genetic features. The preprocessing pipeline includes normalization, encoding, and feature engineering to enhance model performance and ensure consistency in predictions.

To support deployment and scalability, the system is containerized using Docker, which allows the application to run consistently across different environments by packaging all dependencies into isolated containers. Additionally, Nginx is used as a reverse proxy server to manage incoming traffic, improve load balancing, and enhance system security. For data storage and user management, MongoDB is integrated as a NoSQL database, providing flexible schema design and efficient handling of user data. Overall, this combination of technologies ensures that the system is accurate, efficient, scalable, and suitable for real-world deployment.

5 Performance and Evaluation

The performance of the proposed Myopia Risk Prediction System is evaluated using standard metrics suitable for both classification and regression tasks, ensuring a comprehensive assessment of its effectiveness as a screening tool. The system employs a three-stage machine learning pipeline, and each stage is evaluated independently as well as in an integrated manner to analyze overall performance.

In the first stage, which focuses on refractive error detection, a classification model based on XGBoost is utilized. The model achieves a high Area Under the Receiver Operating Characteristic Curve (AUC) score of 0.9431, indicating strong discriminative capability in distinguishing between individuals with and without potential refractive errors. This high AUC value reflects the model's ability to minimize both false positives and false negatives, which is critical in early-stage screening applications.

In the second stage, the system predicts the progression risk level of myopia using another classification model based on XGBoost. The model achieves an overall accuracy of 81.2% along with an AUC score of 0.8842, demonstrating reliable performance in categorizing individuals into low, medium, and high-risk groups. These results indicate that the model effectively captures relationships between lifestyle, demographic, and genetic factors and the likelihood of myopia progression.

The third stage involves the estimation of diopter severity using a regression model based on Gradient Boosting. The model achieves a Mean Absolute Error (MAE) of 1.75 diopters, which indicates that the predicted values are reasonably close to the actual measurements. This level of error is acceptable for screening purposes, where approximate estimation is sufficient rather than exact clinical precision.

To ensure robustness, the models are evaluated using techniques such as cross-validation and train-test splitting, which help minimize overfitting and improve generalization across unseen data. Feature importance analysis is also conducted to identify the most influential factors contributing to myopia risk, providing interpretability and supporting domain understanding.

The performance of different models is illustrated in Figure 1, which includes ROC curves, evaluation metrics comparison, confusion matrix analysis, and feature importance visualization.

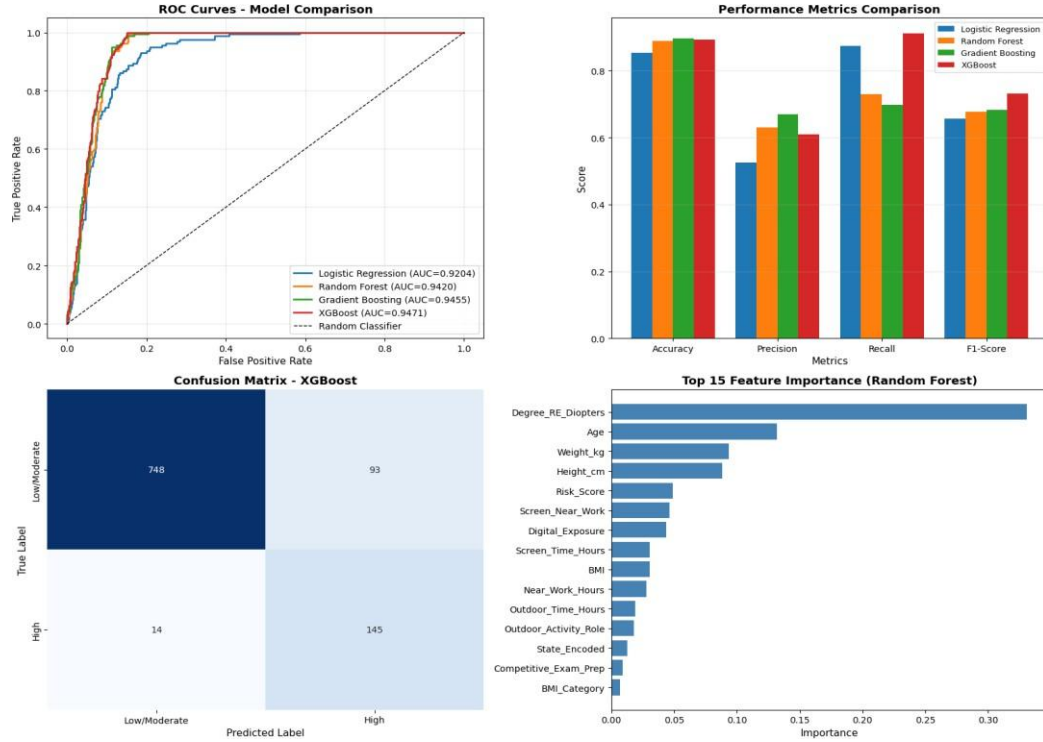


Figure 1: Performance Evaluation of Machine Learning Models including ROC Curves, Metrics Comparison, Confusion Matrix, and Feature Importance.

These results demonstrate that the system achieves highly reliable screening performance, making it suitable for large-scale preventive deployment.

6 Discussion

The proposed system demonstrates strong performance in predicting the likelihood and progression of Myopia using non-invasive input features. The multi-stage pipeline effectively integrates classification and regression models to provide a comprehensive assessment. The first stage, based on XGBoost, accurately identifies individuals with potential refractive error, while the second stage reliably categorizes users into different risk levels. The third stage, implemented using Gradient Boosting, provides a reasonable estimation of diopter severity, enhancing the depth of analysis beyond simple classification. The combination of these stages ensures that the system delivers meaningful and interpretable results suitable for early screening purposes.

In addition to predictive accuracy, the system performs well in terms of usability and real-world applicability. The results are presented through an interactive interface that simplifies complex predictions into understandable visual outputs such as risk levels and percentage scores. This allows non-expert users, including parents and teachers, to

easily interpret the results and take preventive actions. The inclusion of personalized recommendations further strengthens the system by guiding users toward healthier habits, such as reducing screen time and increasing outdoor activities, thereby supporting preventive healthcare.

7 Conclusion

This research successfully outlines and validates the methodology for the proposed Myopia Risk Prediction System. The system presents an effective and scalable solution for the early detection and prevention of myopia using non-invasive and easily accessible input features. By leveraging a multi-stage machine learning pipeline, the system is capable of detecting refractive error, predicting progression risk, and estimating diopter severity with a high degree of reliability. The integration of advanced models such as XGBoost and Gradient Boosting enables the system to capture complex relationships between lifestyle, demographic, and genetic factors, resulting in accurate and meaningful predictions.

In addition to its predictive capabilities, the system emphasizes usability and accessibility through a web-based interface developed using modern technologies such as React and Flask. The ability to provide real-time risk assessment along with personalized recommendations enhances its practical value as a preventive healthcare tool. The system is particularly beneficial for deployment in schools, rural areas, and resource-constrained environments, where access to clinical screening facilities is limited. By enabling early identification of at-risk individuals, the system supports timely intervention and helps reduce the long-term impact of myopia.

Overall, this work demonstrates the potential of artificial intelligence in transforming preventive healthcare by offering a cost-effective, user-friendly, and scalable solution for myopia screening. While the system is not intended to replace professional medical diagnosis, it serves as a valuable first-level screening tool that can significantly improve awareness and early detection. Future enhancements, including the integration of larger datasets and clinical parameters, can further improve the system's accuracy and broaden its applicability across diverse populations.

7.1 Future Performance Optimization

Future work includes improving model accuracy, integrating clinical data, and expanding deployment through mobile applications and multilingual support.

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GitHub: <https://github.com/unnatii14/Myopia-Risk-Prediction>

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